

KENDRIYA VIDYALAYA SANGATHAN ERNAKULAM REGION

ARTIFICIAL INTELLIGENCE (SUBJECT CODE - 417) - CLASS X

PRE-BOARD EXAMINATION 2024-25

TIME: 2 Hours

Max.Marks: 50

MARKING SCHEME

General Instructions

1. Please read the instructions carefully.
2. This Question Paper consists of 21 questions in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. Out of the given $(5 + 16) = 21$ questions, a candidate has to answer $(5 + 10) = 15$ questions in the allotted (maximum) time of 2 hours.
5. All questions of a particular section must be attempted in the correct order.

SECTION A – OBJECTIVE TYPE QUESTIONS (24 MARKS):

- This section has 05 questions.
- Marks allotted are mentioned against each question/part.
- There is no negative marking.
- Do as per the instructions given.

SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):

- This section has 16 questions.
- A candidate has to do 10 questions.
- Do as per the instructions given.
- Marks allotted are mentioned against each question/part.

SECTION A: OBJECTIVE TYPE QUESTIONS

Q.1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
i	Sonia gets up at 6 am and goes for her hobby classes. Then she comes back home and finishes her homework before going to school. She does all work by herself. No one tells her to do so. This is called _____. a. Self-Awareness b. Self-Motivation c. Self-Regulation d. Discipline Ans: b. Self-Motivation	1
ii	_____ act as an interface between the user and the computer. a. Operating System b. MS Excel c. Impress d. Digital Documentation Ans: a. Operating System	1

iii	Assertion (A): An entrepreneur takes risks. Reason (R): An entrepreneur is self-confident. (a) Both A and R are correct and R is the correct explanation of A (b) Both A and R are correct but R is NOT the correct explanation of A (c) A is correct but R is not correct (d) A is not correct but R is correct Ans: Both A and R are correct and R is the correct explanation of A	1
iv	Which of the below is not a step of Effective Time Management? a. Organize b. Prioritize c. Control d. Remember Ans: d) Remember	1
v	Reenu travelled to Spain from India to pursue her higher education. But she doesn't know how to speak Spanish (language of Spain). Because of this, she was unable to find a part time job. This is an example of _____. a. Interpersonal barrier b. Physical barrier c. Organizational barrier d. Linguistic barrier Ans: d. Linguistic barrier	1
vi	Solutions to these global problems like scarcity of clean water to drink, scarcity of pure air to breathe, scarcity of unadulterated food, rising issue of global warming is _____. a. Stop using these resources. b. Sustainability c. to make people aware d. None of the above Ans: b. Sustainability	1
Q2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	Assertion (A): Naturalist Intelligence involves the ability to identify and categories living things and understands other features of the natural world. Reasoning (B): This type of intelligence is exhibited by poets, novelists, and journalists. (a) Both A and R are correct and R is the correct explanation of A (b) Both A and R are correct but R is NOT the correct explanation of A (c) A is correct but R is not correct (d) A is not correct but R is correct Ans: A is correct but R is not correct.	1
ii	Results based on discriminatory assumptions made while developing algorithms by the developer are called as _____. a. AI Ethics b. AI Bias c. Test Data d. Training Data Ans: b. AI Bias	1
iii	Statement A: Neural networks are made up of layers of neurons. Statement B: Human brain consists of millions of neurons. a. Only Statement A is correct b. Only Statement B is correct c. Both the statements are correct d. None of the statements is correct Ans: (c) Both the statements are correct	1
iv	Neural Network is a mesh of multiple _____. Ans: Hidden Layers	1
v	Which of the following is a valid file extension for Notepad file? a. .jpg b. .doc c. .text d. .txt Ans: d	1

vi	are the words which occur very frequently in the corpus but do not add any value to it. a. Special Characters b. Stopwords c. Roman Numbers d. Useless Words Ans: b. Stopwords	1
Q3	Answer any 5 out of the given 6 questions on (1 x 5 = 5 marks)	
i	Which of the following is Not AI? i. Air Conditioner ii. Automatic Washing Machine iii. Self-Driving Car iv. Smart TV a. i, ii & iii b. i & ii c. i, ii & iv d. iii & iv Ans: (c) i, ii & iv	1
ii	Data science is the processing of diverse sets of data through? a. Organizing data b. Processing data c. Analyzing data d. All of the above Ans: d. All of the above	1
iii	Which is one of the significant data science skills? a. Statistics b. Data Visualization c. Machine Learning d. All of the above Ans: d. All of the above	1
iv	Which of the following tasks involves identifying and locating objects within an image? a. Image compression b. Feature extraction c. Object detection d. Image enhancement Ans: c. Object detection	1
v	Identify the type of Chabot with the information given below: These bots work on pre-programmed instructions inside the application/machine and are generally easy to develop. They are deployed in the customer care section of various companies. Their job is to answer some basic queries that they are coded for and connect them to human executives once they are unable to handle the conversation. Ans: Script bot	1
vi.	The percentage of true positive cases versus all the cases where the prediction is true is defined as a. Precision b. Accuracy c. F1 Score d. None of these Ans: a. Precision	1
Q4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	

i	Aaron is facing issues in his internet connectivity at home. He called customer care service; A virtual assistant took his complaint and replied with an acknowledgement number. What is the AI technology used by Internet service providers? a. Data Science b. Machine Learning c. Computer Vision d. Natural Language Processing Ans:d. Natural Language Processing	1
ii	Statement 1: F1 score is evaluated based on precision or recall. Statement 2: When the F1 score is 0, the model accuracy is 100% a. Both statement1 and statement 2 are correct. b. Both statement1 and statement 2 are incorrect. c. Statement 1 is correct but statement 2 is incorrect. d. Statement 1 is incorrect but statement 2 is correct. Ans:c. Statement 1 is correct but statement 2 is incorrect.	1
iii	Accuracy, Recall, Precision, F1 Score are the parameters to calculate the efficiency under _____ . a) Data Acquisition b) Data Evaluation c) Data Testing d) Data Modelling Ans: b) Data Evaluation	1
iv	A grayscale image represents intensity values ranging from 0 to _____ . Ans:255	1
v	What is the process of extracting emotions within a text data using NLP called? a. Sentiment Analysis b. Emotional Data Science c. Emotional Processing d. Emotional Classification Ans: a. Sentiment Analysis	1
vi	An AI system is provided information about the residences, such as their prices, square footage, and number of rooms, features and whether or not they have gardens, using the associated labels. We can now train the model to forecast the price of a new house using data from thousands of houses, their attributes and pricing. Name the illustration. Ans:Supervised learning.	1
Q5	Answer any 5 out of the given 6 questions on Employability Skills (1 x 5 = 5 marks)	
i	A well-known news outlet distributes information on a variety of subjects, such as politics, the federal government, world affairs, sports and the most recent developments in film and television. Gathering, analyzing and disseminating information about current events is the responsibility of reporters. Additionally, the news channel allows viewers to submit breaking news, which the channel's correspondents will verify. However, it could be challenging to tell the difference between true and false news, which could confuse readers and endanger the reputation of a news organization. In order for the audience to have a high level of trust in the news channel, which of the following domains is employed to identify fake news? a. Computer Vision b. Matrices c. NLP d. Data Science Ans: NLP	1

ii	Which evaluation parameter takes all the percentage of correct predictions out of all the observations. Ans: Accuracy	1
iii	Which of the following is a primary color in the RGB color model? A) Yellow B) Cyan C) Green D) Magenta Ans: Green	1
iv	In text normalization, text from multiple documents and the term used for the whole textual data from all the documents altogether is known as _____. a. Corpus b. Tokens c. Lemma d. Stem Ans: Corpus	1
v	Order shipped! Expected delivery in three days! No of tokens in the above sentence is a.10 b.9 c.7 d. 8 Ans: 9	1
vi	An AI model has been developed to detect credit card fraud detections. The aim is not to miss any fraud transactions. Therefore, we want False-Negative to be as low as possible. In these situations, we can compromise with the low precision, but _____ should be high. A. Accuracy B. F1 score C. Recall D. Confusion matrix Ans: c. Recall	1

SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):

Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks)

Answer each question in 20 – 30 words

Q.6	Feedback is the final component and one of the most important factors in the process of communication. Give two reasons to justify why feedback is important. Ans: - It validates effective listening: The person providing the feedback knows they have been understood (or received) and that their feedback provides some value. - It motivates: Feedback can motivate people to build better work relationships and continue the good work that is being appreciated. - It is always there: Every time you speak to a person, we communicate feedback so it is impossible not to provide one. - It boosts learning: Feedback is important to remain focussed on goals, plan better and develop improved products and services. - It improves performance: Feedback can help to form better decisions to improve and increase performance.
Q.7	What is the importance of setting goals in life? Ans: Following are the importance of Goal setting - Goals allow you to separate out what's important. - It helps you to focus on the end result instead of less Important work. - This will make you successful in your Career and personal life.
Q.8	Aisha is searching some contents related to sports in a search engine but during searching help her about how to secure her data while using the internet? Ans: To protect the data following measures should be take care: - Use a strong password in your account. - Install antivirus and firewall - Keep the information in encrypted format - Click only on secure websites during internet surfing.

Q.9	List two qualities of successful Entrepreneurs. Ans. They are confident. They believe in themselves and their abilities. • They keep trying new ideas in their business. • They are patient. • They are creative and think differently about business ideas. • They take responsibility for their actions. (Any two Points)
Q.10	How can we achieve Sustainable development? Ans. Sustainable development can actually happen only when each one of us works towards it. We have to become responsible environment citizens who can protect the environment through our own efforts.

Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)

Q.11	“In the coming years, AI-enabled machines will replace all the people who work as laborers.” Justify this statement. Ans: • Advancements in Technology: AI and robotics are advancing rapidly, allowing machines to perform tasks that were once considered too complex for automation. This includes jobs in manufacturing, agriculture, and even some service industries. • Labor Shortages: In some regions, there is a shortage of available labor. AI and automation can help fill these gaps, allowing industries to maintain productivity without relying solely on human workers.
Q.12	What is targeted advertising? Explain with an example Targeted advertising is a form of online ads tailored to individuals based on their interests, online behavior, and demographics. For example, if you search for "running shoes," you might start seeing ads for sports gear on other websites.
Q.13	Suhana works for a company wherein she was assigned the task of developing a project using AI project cycle. She knew that the first stage was scoping the problem. Help her list the remaining stages that she must go through to develop the project. Ans. Steps of AI project life cycle: 1. Data Acquisition 2. Data Exploration 3. Modelling 4. Evaluation
Q.14	Explain the difference between grayscale and RGB images. Ans: Grayscale images consist of shades of gray, ranging from black to white, with each pixel representing an intensity value. RGB images use three color channels (Red, Green, Blue), where each pixel is a combination of these three colors, allowing for a wide range of colors in the image.

Q.15	Write the sentence segmentation for the following text: “Raj and Vijay are best friends. They play together with other friends. Raj likes to play football but Vijay prefers to play online games. Raj wants to be a footballer. Vijay wants to become an online gamer” Ans: 1. Raj and Vijay are best friends. 2. They play together with other friends. 3. Raj likes to play football but Vijay prefers to play online games. 4. Raj wants to be a footballer. 5. Vijay wants to become an online gamer.													
Q.16	Draw the confusion matrix for the following data • the number of true positive = 150 • the number of true negative =67 • the number of false positive = 72 • the number of false negative = 190 Ans: <table><tr><th colspan="2" rowspan="2"><i>Confusion Matrix</i></th><th colspan="2"><i>Reality</i></th></tr><tr><th><i>Yes</i></th><th><i>No</i></th></tr><tr><th rowspan="2"><i>Prediction</i></th><th><i>Yes</i></th><td>150</td><td>72</td></tr><tr><th><i>No</i></th><td>190</td><td>67</td></tr></table>	<i>Confusion Matrix</i>		<i>Reality</i>		<i>Yes</i>	<i>No</i>	<i>Prediction</i>	<i>Yes</i>	150	72	<i>No</i>	190	67
<i>Confusion Matrix</i>				<i>Reality</i>										
		<i>Yes</i>	<i>No</i>											
<i>Prediction</i>	<i>Yes</i>	150	72											
	<i>No</i>	190	67											

Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)

Q.17	After class 12 Rahul wanted to join for AI course. His parents didn't know much about its domains. Explain to them the domains of AI. Ans: Domains of AI: Data Science: It involves algorithms that enable computers to learn from data and make predictions or decisions. Natural Language Processing (NLP): It focuses on enabling computers to understand, interpret, and generate human language. Computer Vision: This domain enables computers to interpret and analyze visual information from the real world, such as images or videos. Robotics: It involves the design and creation of robots capable of performing tasks autonomously or with human assistance.
Q.18	Draw the 4Ws problem canvas and explain each one of them briefly. Ans: The 4Ws problem canvas is the basic template while scoping a problem and using this canvas, the picture becomes clearer while we are working to solve it. a) Who: The “Who” block helps you in analyzing the people getting affected directly or indirectly due to it? Under this, you find out who the ‘stakeholders’ to this problem are and what you know about them. Stakeholders are the people who face this problem and would be benefited with the solution. b) What: Under the “What” block, you need to look into what you have on hand. At this stage, you need to determine the nature of the problem. What is the problem and how do you know that it is a problem? c) Where: In this block, you need to focus on the context/situation/location of the problem. It will help you look into the situation in which the problem arises, the context of it, and the locations where it is prominent. d) Why: in the “Why” canvas, think about the benefits which the stakeholders would get from the solution and how it would benefit them as well as the society.

Q.19	Differentiate between rule-based and learning-based AI modelling approaches. Ans: Rule Based Approach: It refers to the AI modelling where the relationship or patterns in data are defined by the developer. The machine follows the rules or instructions mentioned by the developer, and performs its task accordingly. For example, suppose you have a dataset comprising of 100 images of apples and 100 images of bananas. To train your machine, you feed this data into the machine and label each image as either apple or banana. Now if you test the machine with the image of an apple, it will compare the image with the trained data and according to the labels of trained images, it will identify the test image as an apple. This is known as a Rule based approach. The rules given to the machine in this example are the labels given to the machine for each image in the training dataset. Learning Based Approach: In this approach, the machine learns by itself. It refers to the AI modelling where the relationship or patterns in data are not defined by the developer. In this approach, random data is fed to the machine and it is left on the machine to figure out patterns and trends out of it. Generally, this approach is followed when the data is unlabelled and too random for a human to make sense out of it. For example, suppose you have a dataset of 1000 images of random stray dogs of your area. You would put this into a learning approach-based AI machine and the machine would come up with various patterns it has observed in the features of these 1000 images which you might not have even thought of!
Q.20	Consider the text of following documents: Document 1 : Sahil likes to play cricket Document 2 : Sajal likes cricket too Document 3: Sajal also likes to play basketball Apply all four steps of the bag of words of NLP on the above given documents and generate output. Ans: Step 1: Tokenization We break down each document into individual words (tokens). - Document 1: ["Sahil", "likes", "to", "play", "cricket"] - Document 2: ["Sajal", "likes", "cricket", "too"] - Document 3: ["Sajal", "also", "likes", "to", "play", "basketball"] Step 2: Vocabulary Creation We create a list of unique words from all documents. Vocabulary: ["Sahil", "Sajal", "likes", "to", "play", "cricket", "too", "also", "basketball"] Step 3: Encoding We create a feature vector for each document based on the vocabulary. Each document will be represented as a vector where the index corresponds to the word in the vocabulary. - Document 1 Vector: [1, 0, 1, 1, 1, 1, 0, 0, 0] (Sahil=1, Sajal=0, likes=1, to=1, play=1, cricket=1, too=0, also=0, basketball=0) - Document 2 Vector: [0, 1, 1, 0, 0, 1, 1, 0, 0] (Sahil=0, Sajal=1, likes=1, to=0, play=0, cricket=1, too=1, also=0, basketball=0) - Document 3 Vector: [0, 1, 1, 1, 1, 0, 0, 1, 0] (Sahil=0, Sajal=1, likes=1, to=1, play=1, cricket=0, too=0, also=1, basketball=1) Step 4: Output Generation The final output consists of the vocabulary and the encoded vectors for each document. Vocabulary: ["Sahil", "Sajal", "likes", "to", "play", "cricket", "too", "also", "basketball"] Document Vectors: - Document 1: [1, 0, 1, 1, 1, 1, 0, 0, 0] - Document 2: [0, 1, 1, 0, 0, 1, 1, 0, 0] - Document 3: [0, 1, 1, 1, 1, 0, 0, 1, 0]

Q.21	A social media company has developed an AI model to predict which users are likely to churn (cancel their account). During testing, the AI model came up with the following predictions.			
	Confusion Matrix	Reality	Yes	No
	Predicted	Yes	60	25
		No	5	10
	(i) Calculate precision, recall and F1 Score for churn prediction (ii) How many total tests were performed in the above scenario? Ans: (i) Precision = True Positives / (True Positives + False Positives) = $60 / (60 + 5) = 0.923$ Recall = True Positives / (True Positives + False Negatives) = $60 / (60 + 25) = 0.706$ F1 Score = $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}) = 2 * (0.923 * 0.706) / (0.923 + 0.706) \approx 0.801$ (ii) Total tests performed = Sum of all entries in the confusion matrix = $60 + 25 + 5 + 10 = 100$.			

